NIHVA Technologies Pvt Ltd



We Make Possible

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Clean Room Comparison NIHVA ASSA ABLOY

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

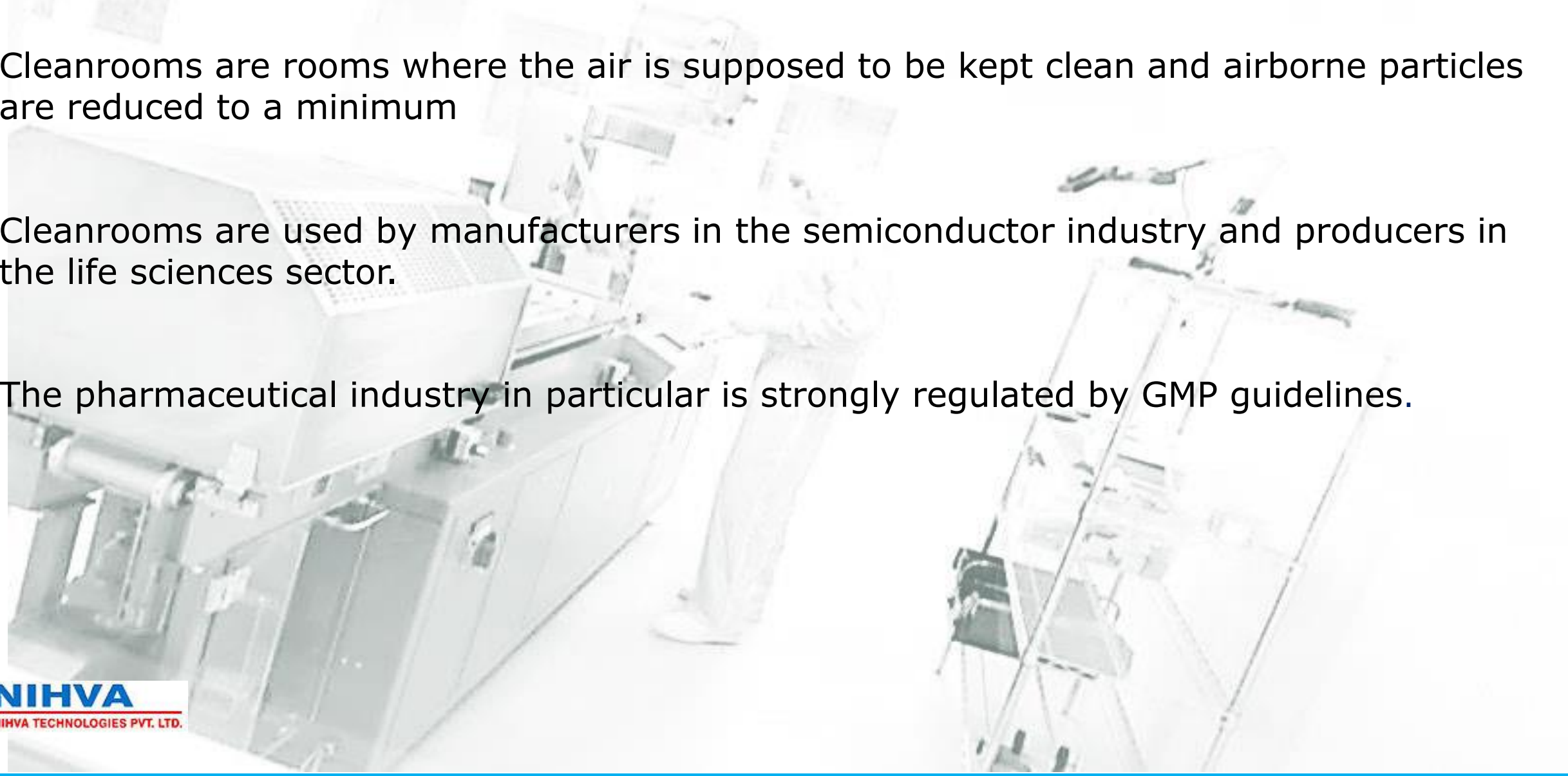

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Where is cleanroom technology used?

Cleanrooms are rooms where the air is supposed to be kept clean and airborne particles are reduced to a minimum

Cleanrooms are used by manufacturers in the semiconductor industry and producers in the life sciences sector.

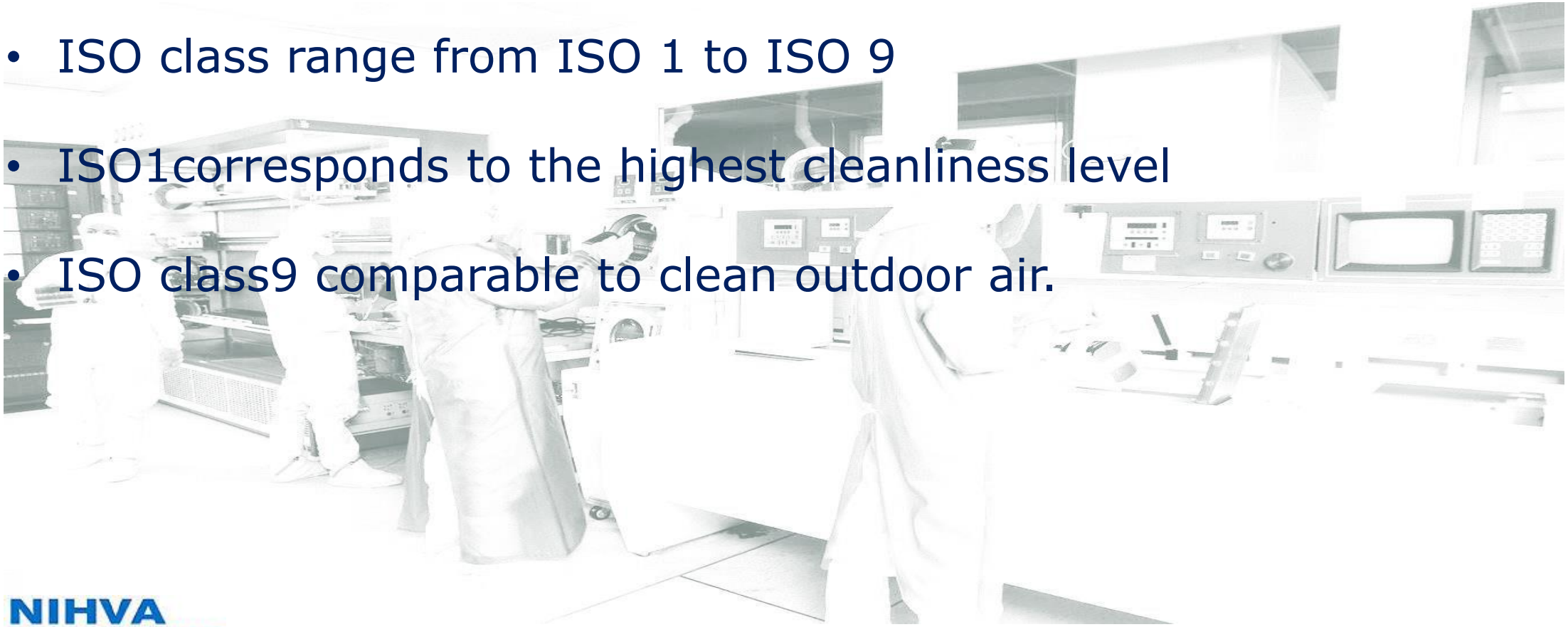
The pharmaceutical industry in particular is strongly regulated by GMP guidelines.



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

What are cleanroom classes and how are they defined?

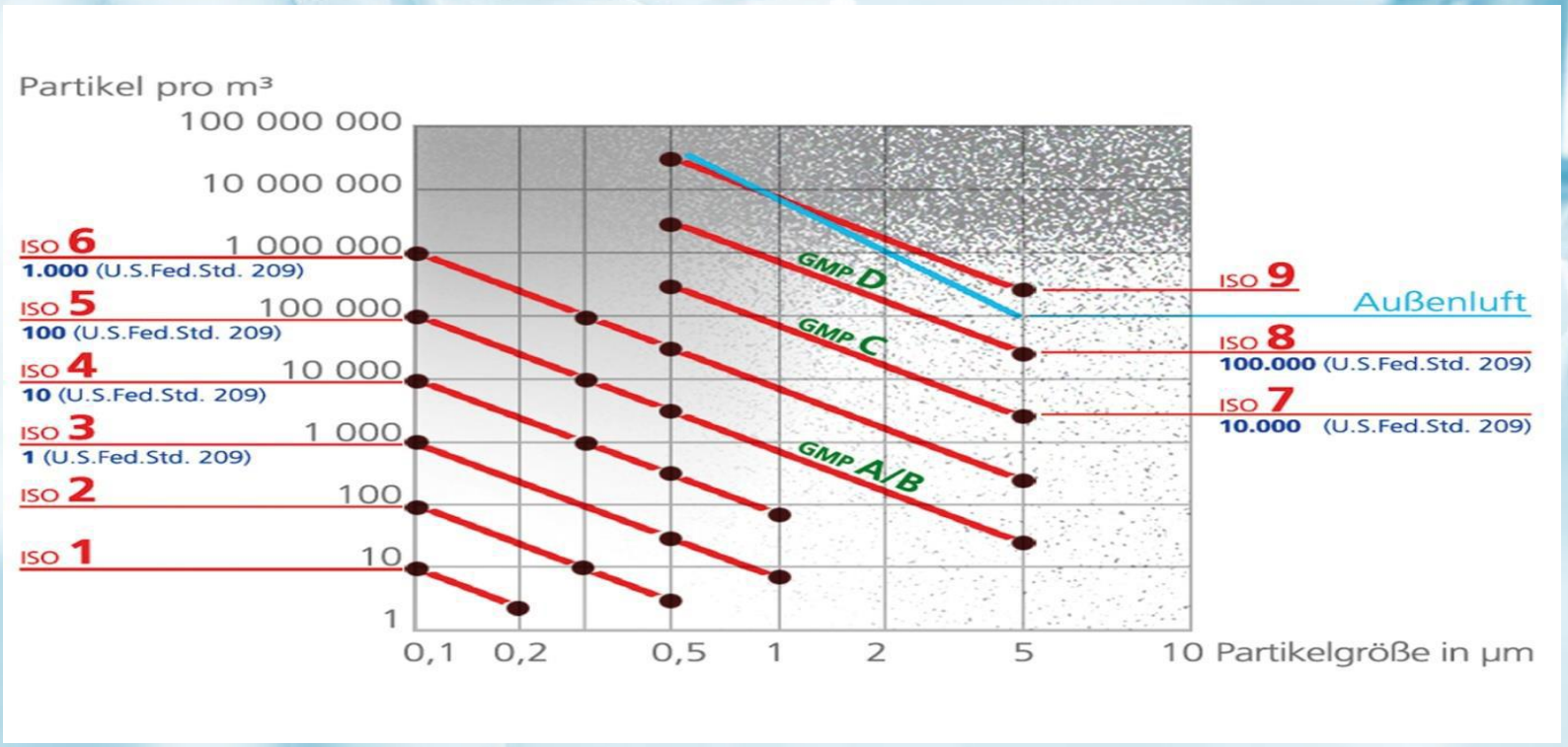
- EN ISO 14644-1 standard defines clean room classes according to the cleanliness level of clean room
- ISO class range from ISO 1 to ISO 9
- ISO1corresponds to the highest cleanliness level
- ISO class9 comparable to clean outdoor air.



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

What is a ISO concept?

- The more critical the product or the process, the lower the required ISO class
- Requirements on particle and microorganism concentrations, room air conditioning and air change rate, surfaces of rooms and equipment, work clothing, monitoring, and documentation apply to each ISO class.



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

What is the allowable particle concentration?

Cleanroom class	Shutdown		Normal operation	
	Maximum allowable number of particles per m^3 (equal to or greater than)			
	0.5 µm	5.0 µm	0.5 µm	5.0 µm
ISO 5 = A	3,520	20	3,520	20
ISO 6 = B	3,520	29	352,000	2,900
ISO 7 = C	352,000	2,900	3,520,000	29,000
ISO 8 = D	3,520,000	29,000	Not defined	Not defined



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

What is a GMP zoning concept?

- Good manufacturing practice (GMP)
- Guidance for manufacturing, testing, and quality assurance
- Minimum requirements to assure drug product is safe for human consumption
- GMP follows principles:
  - Hygiene
  - Controlled environmental conditions
  - Controlled manufacturing processes
  - Documentation…
- Target requirement for air change rate and particle concentration.
- Many products must be produced in designated hygiene zones with different cleanliness classes
- RR300 Clean certified as suitable product for “GMP Class C” areas

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

What is a GMP zoning concept?

GMP class	Application
A	Zones for high-risk operations
B	Background environment for a cleanliness class A zone for use in aseptic production
C	Zones for less critical stages in the production of sterile products
D	Zones for less critical stages in the production of sterile products

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Clean Room Comparison NIHVA ASSA ABLOY



HS9020GAT



HS9030GAT



Rapid Roll Clean

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

SR. No	ASSA ABLOY	HS9020GAT	HS9030GAT	Rappid Roll 300 Clean
1	Operation	Gravity	Gravity	Electrical 0.75 kW
2	Min Size	1000 x 2000	1000 x 2000	1000 X 1000
3	Max Size	4000 x 4000	4000 x 4000	3500 X 3500
4	Opening Speed	Up to 1,2 m/s , optional 2,0 m/s	Up to 2,7 m/s	Up to 2,0 m/s
5	Closing Speed	0,5 m/s	0,5 m/s	1 m/s
6	Construction	Standard Galvanised Steel	Standard Galvanised Steel	Standard Stainless Steel
7		Option: Stainless Steel 304	Option: Stainless Steel 304	Option: Galvanised steel or powder steel
8	Covers	Option: Galvanised steel or powder steel	Standard Galas fiber reinforced plastic	Option: Galvanised steel or powder steel
9	Control Box	Exposed Control Box: Additional Space required cables not integrated	Exposed Control Box: Additional Space required cables not integrated	MCC control integrated clear text display with error log integrated in side frame
10	Wind Load	22 m/s - class 1 In operation : 9 m/s -50 Pa	22 m/s - class 1 In operation : 9 m/s -50 Pa	22 m/s - class 1 In operation : 9 m/s -50 Pa
11	Air Permeability	Class 3/1	Class 4 Overpres Class 3 underpres	21 m^2 /h at pressure diff 25 Pa for door DW 1500 X DH 2000mm

Sr. No	ASSA ABLOY	HS9020GAT	HS9030GAT	Rapid Roll 300 Clean
12	Side Space	65	81	110
13	Head Room	390	402	470
14	Depth	255	328	340
15	Placement	Right / left	Right (Inside the drum)	Right / left
16	Certification	ISO Class 7	ISO Class 6 - EN 14644-1 Certified GMP Class C Certified	ISO Class 5 - EN 14644 -1 Certified GMP Class C Certified
17	Safety	Wireless safety edge & Stationary photocell	Wireless safety edge & Stationary photocell	Contact safety edge & Stationary photocell Optional : Touchless safety edge
18	Bottom beam	Flexible bottom beam reduce corridor space. Movement back and forth cause scratches	Flexible bottom beam reduce corridor space. Movement back and forth cause scratches	Right stainless steel bottom beam with rubber Profile sealing
19	UPS	No Integrated System	No Integrated System	Integrated Self-reloading UPS system for more than 30 opration , all function remain
20	Curtain	Zipper curtain becomes unstable after some crashes .curtain will not remain in side columns when pressure is affecting it.	Zipper curtain becomes unstable after some crashes. Curtain will not remain in side columns when pressure is affecting it.	Rigid bottom bar and wind knobs on the curtain Keep door curtain in side frame guides
21	Activators	Activator or light can only be place outside the door	Activator or light can only be place outside the door	Activators and LED-lights can be integrated in columns

Clean Room Price Comparison NIHVA ASSA ABLOY

Supply + Installation + Interlocking Price

Sr. No	Size : (W) x (H) in mm	Qty	HS9020GAT Price in INR (Each)	Total Price in INR	HS9030GAT Price in INR (Each)	Total Price in INR	RR300 Clean Price in INR (Each)	Total Price in INR
1	1050 x 2200	2						
2	1500 x 2200	2						
	Total	4		1,86,84,000		2,52,11,000		2,74,56,000

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.


HERMES
PHARMA


BAYER


Johnson
Controls


ABBOTT


PALL
Pall Corporation


SALUTAS Pharma GmbH


Dr. Ing. Huber


MERCK


haupt
Pharma


gsk
GlaxoSmithKline


PERRIGO


Pfizer


WACKER


ZEISS


NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA – Major Client List

Automobile

TATA MOTORS, Mercedes-Benz, Mahindra, FIAT, NISSAN, BAJAJ

Industrial

SANDVIK, PARKER, EPCOAT, BEKAERT, eisenmann, FINEARC SYSTEMS, KONECRANES, EATON, faurecia, ATS Group, SKF, PMP INDUSTRIES, StanleyBlack&Decker, MAGNA COSMA INTERNATIONAL, FUKOKU, SIEMENS, Atlas Copco, ROSS, SIGNODE, TAKENAKA INDIA, JOLLY, TATA TOYO, WF, SULZER, GIFT, ALSTOM, RAWAT, IR, GE, WILO, Mather + Platt, TATA, adani, ZF INDIA, BOSCH, Invented for life, DUPONT, BASF, SEIL, SKIPPER, RELIANCE

Food

Puratos, COLGATE-PALMOLIVE, FERRERO, Haldiram's, INBISCO, BARRY CALLEBAUT, Nestle, MARS

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Pharmaceuticals:

LUPIN
ajanta pharma limited
Abbott
A Promise for Life
Emcure
Success through Innovation

Contractor / PEB

Shapoorji Pallonji
SAMARTH GROUP
Kalbhor Group
REWARD
KHAMKAR CONSTRUCTIONS PVT. LTD.
PHOENIX INFRA
TRUTH IS OUR ORNAMENT
PHOENIX INFRA ESTATE INTERNATIONAL LTD.
ZAMIL STEEL
IVRCL
ROHAN
M/s.Sandip Nanavati
Engineers & Contractors
ADITYA CONSTRUCTIONS
SHMZ
mb Chitale Constructions
NYATI GROUP

Architects / PMC

CRN
vms
Venkataramanan Associates
ARCHVISTA
ARCHICON
Simplifying Spaces
snpc
SHAH & TALATI
Engineering & Architecture
JLL
synefra
SMPS
CONSULTANTS TO INDUSTRY
ICL
PHENIX
Creative Simplified
Knight Frank

Jewellery & Fashion

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Thank You

International Business Partners

ALBANY ASSA ABLOY

MEGADOOR ASSA ABLOY

Crawford ASSA ABLOY

besam ASSA ABLOY

ACO

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.